

# HR Analytics Dashboard

Project Report | Employee Attrition & Workforce Analysis

A concise report of the dataset, dashboard metrics, workforce patterns, and practical HR analysis opportunities.

| Dataset Profile       | Value |
|-----------------------|-------|
| Rows                  | 1,480 |
| Columns               | 38    |
| Missing cells         | 57    |
| Exact duplicate rows  | 7     |
| Attrition = Yes (CSV) | 238   |
| Attrition = No (CSV)  | 1,242 |

Dataset attrition rate calculated from the CSV: 16.1%. The dashboard screenshot displays 1.416K employees, 229 attrition, and a 16.2% attrition rate; dashboard KPIs may reflect the Power BI model's measures and filters.

# 1. Overview

This project uses HR data to explore employee attrition and workforce characteristics. The Power BI dashboard brings together headline metrics and breakdowns by gender, education, age, job role, salary band, and year. The purpose is to help HR stakeholders spot patterns and decide where deeper investigation may be useful.

# 2. Dataset

The provided HR\_Analytics CSV contains 1,480 records and 38 columns. Relevant fields include employee demographics, department, job role, education, monthly income, years at company, and attrition status.

# 3. Tools

- CSV dataset for source data
- Microsoft Power BI for dashboarding and visualization
- Power Query for data preparation, where applicable
- DAX measures for KPI calculations, where applicable

# 4. Analysis Workflow

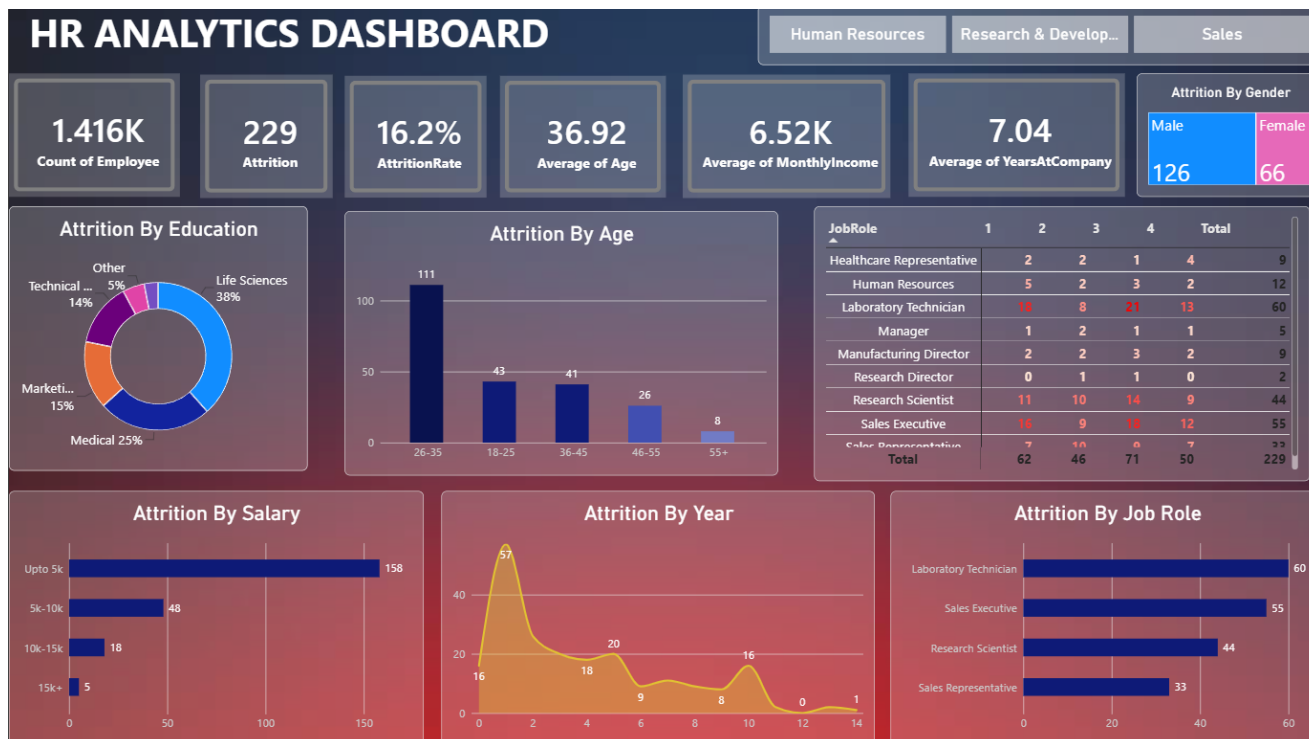
- 1. Load data:** Import the CSV into Power BI and review the available columns.
- 2. Inspect data:** Check field types, missing values, duplicates, and category consistency.
- 3. Prepare data:** Correct data types and transform fields where required for accurate analysis.
- 4. Define metrics:** Review employee count, attrition count, attrition rate, average age, monthly income, and tenure measures.
- 5. Build visuals:** Use KPI cards, charts, and a job-role/level matrix to compare employee attrition patterns.
- 6. Review findings:** Interpret the dashboard and identify questions for HR follow-up.

# 5. Dashboard Overview

The supplied dashboard screenshot shows the following headline metrics:

| Dashboard KPI            | Displayed value |
|--------------------------|-----------------|
| Count of Employee        | 1.416K          |
| Attrition                | 229             |
| Attrition Rate           | 16.2%           |
| Average Age              | 36.92           |
| Average Monthly Income   | 6.52K           |
| Average Years at Company | 7.04            |

## Dashboard Preview



The screenshot includes attrition breakdowns by gender, education, age, salary band, year, and job role, plus a job-role table by job level.

## 6. Results & Observations

The CSV contains 238 records marked as attrition = Yes and 1,242 marked as No. The dashboard screenshot reports 229 attrition cases and a 16.2% attrition rate. Differences between CSV-derived counts and dashboard measures should be checked in the Power BI model before publication.

The dashboard visually indicates that attrition is concentrated in the lower salary bands and that Laboratory Technicians and Sales Executives are among the largest job-role groups in the displayed attrition chart. The age chart shows the largest displayed group in the 26–35 range. These are descriptive observations from the screenshot, not proof of causes.

## 7. Potential HR Actions

- Investigate attrition patterns in lower salary bands alongside role, tenure, and satisfaction data.
- Review retention and career-development options for job roles with higher attrition counts.
- Compare attrition by age, education, and department to identify where additional analysis is needed.
- Validate all KPIs and definitions before sharing the dashboard with stakeholders.

## 8. How to Run

1. Keep *HR\_Analytics(1).csv* and the Power BI workbook in an accessible folder.
2. Open the HR Analytics dashboard in Power BI Desktop.
3. If prompted, update the data source path to the provided CSV file.
4. Refresh the data and verify the employee count, attrition measure, and rate.
5. Review the dashboard using department tabs and the available charts.
6. Save a copy after confirming that the visuals and measures update correctly.

## 9. Project Files

- *HR\_Analytics(1).csv* — source dataset
- *HR\_Analytics(1).pbix* — Power BI dashboard
- *HR\_Analytics\_Report.pdf* — project documentation

### Skills Demonstrated

HR analytics • Data profiling • Power BI • KPI reporting • Data visualization • Attrition analysis • Business interpretation

